



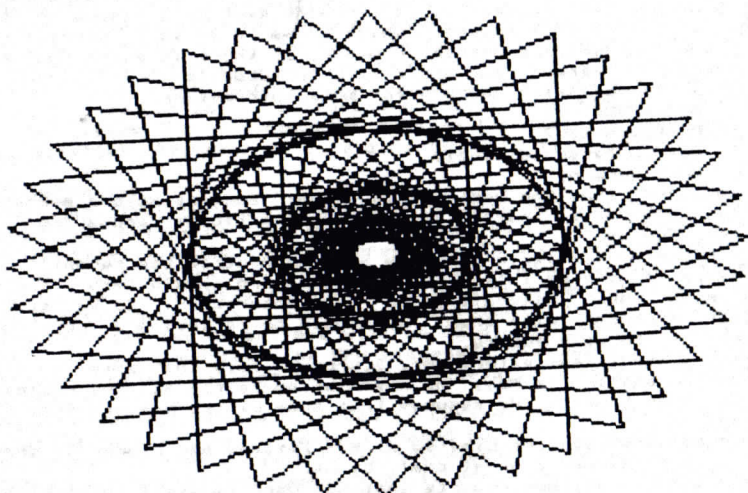
de Montréal

(514) 455-4705

443 Boulevard Roche. Vaudreuil. Québec. Canada. J7V 2M4

Vol.:2 No.:11

SEPTEMBER/SEPTEMBRE 1986



If the September meeting goes good, we will have our next meeting the first of October at 7:30 pm, downtown 3460 Peel St. suite 111.

NEWSLETTER ARTICLES

Members are encouraged to contribute to the Newsletter in the form of articles, mini programs, helpful tips, jokes, cartoons and questions. A short article may be submitted in any form in English or French. We welcome the reprinting of any article appearing in this newsletter providing credit is given to the author and to SMUG of Montreal. Opinions expressed in this newsletter are those of the writers and are not necessarily those of SMUG of Montreal. SMUG of Montreal cannot assume liability for errors in articles, programs or advertisements.

SMUG NEWS

Mr. Uwe Schroeder of the West Germany SVI/MSX User Group has informed us that the West German SVI distributor has dropped the SVI-318 & 328 computers. We would like to thank Mr. Schroeder for the SV-BASIC public domain programs that he sent to us. He also sent us 6 issues of the German MSX magazine called "MSX Revue", which seems to be very interesting, even though I don't understand German. However, if there is anyone out there who does understand German and would like to subscribe to it, you may by writing to the following address for more information:

MSX REVUE
Postfach 1107
8044 Unterschleissheim
West Germany

More news from the West German group! Douglas Boyd a member of the group has offered to translate some of the articles appearing in their newsletter, for us.

Mr. Wilhelm Pitzeier Jr. of the SV Club Austria has recently contacted us and asked if we'd be interested in exchanging newsletters, information, etc... with them. Naturally, we are interested! So we started the newsletter exchange with them. We should be receiving a copy of their newsletter in the near future.

Talking about newsletters! Mr. Schroeder said that after he held their first group meeting in May, he was missing nearly all the copies of our newsletter that he had. He assumes that some of the members from his group took them.

Computer Distributors Limited of New Zealand has closed up shop. Their support will be missed a great deal, by us.

We have unconfirmed reports that MSX User is still publishing their magazine. So I would suggest that those members who subscribed to the magazine last year, but, never received an issue or a refund write to them at the following address and get on their case about it:

ARGUS SPECIALIST PUBLICATIONS LIMITED
No. 1 Golden Square
London W1R 3AB
England

If you have no luck with Argus, you may write to the following agency who may be able to help you, to either get your money back or to get a year's worth of issues from MSX User:

THE ADVERTISING STANDARDS AUTHORITY
Dept 2, Brook House
Torrington Place, London WC1E 7HN
England

After all it's your money that they have and used for a whole year interest free and not giving you anything in exchange. So what's a few minutes of your time to write a letter of complaint and the 68 cents postage, compared to the \$30.00 Canadian plus unpaid interest, that you've already invested?

THE SYMBIOTIC COMPUTER CONCEPT

By: John Durham

(An extension of the way in which modern computers work)

DESCRIPTION:

Most present day computer systems are designed around the concept of a single processing unit. Small systems use a single microprocessor, while large ones may vary from single large processors to many small processors operating around a global memory store. In most systems, extra buffers are supplied to protect the system from heavy system fanout loads.

The first step in altering this arrangement is to introduce a second set of buffers into the region between the CPU and its main buffers. The original set of buffers will still perform its usual function of driving the address and data lines which control the main (internal) system memory, ports etc. The second set of buffers are used in the process of taking control of the system bus of a computer outside the host for long enough to carry out parallel transfer of data directly into the outside system's memory.

THE CONTROL MECHANISM:

This is controlled by a port, which is located between the CPU and its main buffers. The purpose of this port is to divide up the available memory into four identical windows, each represented by a bit on the port. The state of each bit determines whether a memory window exists inside or outside the system. All these bits are set to 0 on power up, ensuring that during the bootstrap process, all memory is internal to the system, and only the conventional set of buffers will be used. If any of these bits is set to 1 at any time, and then the CPU addresses that part of memory, the second set of buffers is used to request access to memory outside the system for that operation.

During external operations, access is controlled by a bus request and bus acknowledge system, which allows unused bus lines in the external system to be handed over briefly to this system. The CPU then uses the external memory in exactly the same way as it uses its own. This means that large blocks of data can be moved from the memory of one system to that of another using direct block shift instructions, such as LDIR for the Z-80. The CPU is oblivious of this process because it still appears to be working within its own memory space.

The external operations themselves are controlled by a gate array, which uses the data available on the internal port to decide which set of buffers will be used, or if RAM or ROM masking is to take place. A bit is also provided on the internal port to control whether the bootstrap ROM is switched in or out of operation, and also one controls whether the logical ports are inside or outside the system (it can use its host's peripherals as its own if required).

In this configuration, the main system, and its external (host) system exist in a symbiotic relationship in which each can carry out its own

operations, but data can be passed backwards and forwards under control of the main system, directly from one memory to the other. Thus, by devising a protocol for system communication, the host system can leave messages to be picked up by the main system, and the main system can write messages directly to the host.

BUS ARBITRATION:

By definition, the logic gate array arbitrates between the main system and host's use of the host's memory bus system. Since the two bus systems are separate, however, between the internal and external buffers, the logic may grant access to the main system's internal bus, while the CPU is busy accessing the external one. In this way, the main system can become the host to another identical machine, connected to its own internal bus and memory systems. Likewise, by conventional bus requesting and acknowledging, the main system can grant access to another unit which is using it as a host. If the memory bus connectors are duplicated on the internal bus side of the main system, then two other systems may use the main system as a host. By doing this small piece of duplication, a computer unit is formed which can expand system by system to form a binary tree of CPU systems extending almost to infinity.

BRAIN MIMICKING:

The reason that systems are designed in this way has to do with the differences between present day computer systems and the human brain. Computers presently rely on one processor doing every required operation sequentially, as defined by its program. The human brain, on the other hand, while far slower in operation, can handle much more complex problems, using a great deal more information, with blinding speed. This is because the brain has thousands of processing channels operating on the same problem at the same time, and enough spare capacity to work on other things as well. This computer concept, by extending into a binary tree of enormous size, provides a type of simulation of this effect by allowing many CPU's to work together or separately as circumstances require.

COMPARISON WITH CONVENTIONAL COMPUTER SYSTEMS:

Working with a system like this would have a number of advantages and disadvantages:

ADVANTAGES:

1: All units in the system can be virtually identical, except for peripherals, making them easy to design and cheap to manufacture.

2: Transferring of messages by parallel memory block shifting is just about the fastest possible way of doing so. The machine would therefore be very fast.

3: They could be made with an identical standard bus, which would make all of their peripherals interchangeable.

4: Almost any CPU capable of using ports could be used in this system.

5: The system expands virtually to infinity.

6: The system is capable of handling high rate multitasking applications because of its many CPUs and parallel data transfer capability.

7: The symbiotic relationship between local units which gives rise to the concept's name, allows each CPU to operate with twice its normal capacity and compliment of ports and peripherals.

8: This system would be capable of controlling human like activities, such as hearing, speech, vision, movement, high level decision making and internal housekeeping all simultaneously.

9: It's ideal for use in android style and conventional

industrial robots.

10: Because its bus can be made to match that of a home computer, it can be programmed using one, which it treats as being its initial host. Conventional computer languages can be used.

11: Units can be built with existing technology. No ultra high speed or special components need be used.

12: Each CPU deals with only one host and a maximum of two other peripheral units which pick up messages from it or deposit messages in it.

DISADVANTAGES:

1: The relatively small memory size per CPU, while maximising the efficiency of the unit, makes large database operations more difficult to handle because of the limited amount of information each CPU can control. Limitations of this sort, however, can be solved by alphabetic listing of data base entries, and apportioning different parts of large listings to different CPU units. Thus, it could be made to handle data bases and list searching very rapidly.

2: The potential of this system to expand could lead to large systems requiring multiple or large power supplies. This problem also exists for conventional computers of similar sizes. This problem could be eased by the use of CMOS or HCMOS parts in construction, which require far less power than LSTTL.

3: Multiplicity of boards in such systems could lead to large volume systems, requiring too much room to be mobile as in robots. This problem could be eased by designing boards with several computer units on the one layer, thus magnifying the power and reducing the noise problems such a system would experience.

4: Parallel data transfer at tremendous speeds would be taking place across ribbon cables and through bus connectors. Choice of high reliability bus connectors, shielded cables, good quality buffer chips, reduced knitting of tracks and noise reduction capacitors would be essential considerations during design. This function also requires that ribbon cables be short, as their reliability decreases with the inverse square of the distance from the signal source.

5: Long distance communications would still be along serial lines, allowing in some cases for serial RS232 or current loop links to be set up. This, however, would not be essential in all cases, as many systems would be capable of operating without line data, or of acquiring data from other sources.

6: Some groups, such as business users might be inclined to criticise this type of development because it does not have immediately obvious applications in business or moneymaking operations. Business applications are well catered for by modern computer systems, and this system should not be seen as the be all and end all of computer development. It is for specialized applications, suitable for intelligent systems, heuristic learning, robotics and general applications requiring thinking machines, not business.

ENVIRONMENT:

To the CPU, there is no distinction between memory that is inside and that which is outside its own system. There is no distinction between its own port controlled peripherals and those of its immediate neighbour. If the system has programs to run them, it can operate both equally well. Some peripherals, which interrupt their host, could be programmed to do so by an outside machine, in order to announce arriving messages, for instance.

Operating system ROMs could be switched in or out, maximising use of the available RAM memory. It is suggested that the memory under direct

control of each CPU is divided into two blocks. One block may be used for the operating system, windowed for exchange of data between systems and general housekeeping operations. The other half of the memory is then set aside for user programs which can then be run without interference or forced crash by inappropriate accessing by other systems.

AUTONOMOUS OPERATING SYSTEMS:

This is a term invented for use with this type of machine. It describes a machine code operating system which powers up and tests each unit, keeps the network of CPU systems under control, handles transfer of all data between systems, and carries out program acquisition, memory bank control, window control, peripheral control, intelligent distribution of work and other clever and invisible tasks that the system would be required to perform. Such systems are extremely hardware dependant, and would have to take account of the microcomputer host system and programs running on it. To some extent it would have to emulate the starting system, so that programs already developed could be run without modifications wherever possible.

For example, suppose a unit was connected to a System 80 micro as initial host. The operating system could opt to switch the lower 16k of memory to external operation, thus making the System 80's ROM, keyboard and screen its own, and then boot the System 80 operating system from its disk drives, access a program and switch to internal operation again, releasing the System 80 for normal use. During this time, the System 80 could be running its own program quite unaware that its hardware was being used, as long as the program did not require duplicate access to any of the IO peripherals being operated by the external system. The external system could be programmed to temporarily divert routines for these peripherals into harmless loops while accessing is being carried out, and restore them when accessing is finished.

Anyone interested in discussing this development is invited to write to:

JOHN DURHAM
16 Hudson Ave.
Upper Hutt
New Zealand

THE MSX STANDARD - THE NEW COMPUTERS By: Robert Chapman Wood

Reviewed by: R.L. Caris, member of SMUG de Montreal

A new book that details the MSX story from it's beginnings to what we might expect computers to be doing in our homes in 1995! I found the book very interesting as the term "MSX" is relatively new to me, I found the background information fascinating.

The author discusses the latest version, MSX-2, which was introduced in Japan in the summer of 1985. This version has features that the author termed as "superb graphics, can expand up to 1 megabyte of RAM plus 128k dedicated to video display. The machine can also display 40 or 80 character lines of text, built in clock and calendar with a battery backup, graphics superimpose system, support for a light pen and optional MSX Audio Chip which can mimic all sorts of non-computer sounds."

The author also covers the relationships between Microsoft, ASCII Corporation, and the various supporting companies, mostly in Japan. Various chapters cover the workings of the MSX machines, BASIC, graphics, disk drives, music and so on. There is a substantial amount of information about the interfacing of the computer to various other peripherals. I found the information on interfacing to be most interesting, and in particular the

information on disk drives. The author states that the MSX-DOS commands are the same as MS-DOS, and that an MSX with the 5 1/4" drive can read a data file created on an IBM or compatible, and vice versa!! (I wonder, has anyone tried this yet??) If this is true, then the MSX can read BASIC, MSX-DOS and data files created under MS-DOS (the SV-738 can also use CP/M). This would indeed make the MSX a very remarkable machine! (I note that IBM's new laptop uses the smaller disk as is used in the MSX).

I found the book to contain considerable information that will help me in using my SVI, but provides ideas for other possible uses that I had not even thought about, a good book for my library!!

Published by TAB BOOKS Inc., Blue Ridge Summit, Pennsylvania, 17214
ISBN 0-8306-09075-5, 1985...

PANDA SOFT'S "WORD GAME" REVIEWED

By: Frances Fontaine

This piece of software contains two word games that are rather popular.

One is called CONVICT, which is actually a very good version of the hangman game. It is not easy to guess the words, even though the first and last letters in the word are displayed. In this version of the game the user may either guess one letter at a time or take a chance and guess the entire word. If you should happen to want to guess the whole word at once and you get it wrong you'll lose 2 points. However, if you're right, you don't waste any of the points that you may have left, as you would if you guessed individual letters and maybe got one or two wrong as you progressed in the game. The graphics in CONVICT are excellent and the game is very well done.

The other game is a word scramble game called HUSTLE. In this game, a word consisting of 10 letters is displayed in a scrambled fashion on the screen. Your job is to unscramble it, before the computer does. Very difficult to do, even for an avid "SCRABBLE" player like me.

For more information about WORD see catalogue for details.

CLASSIFIED ADS

FOR SALE: 1 SV-328, 1 SV-904 cassette recorder, 1 80 green screen monitor, 2 SV-902 disk drives, 1 SV-601 expander, 1 80 column card, 1 SV printer interface card and small SV printer, 1 RS232 card, 1 disk controller card, 1 300 baud modem, 1 Coleco adapter & 5 coleco cartridges, 2 Coleco super joysticks and disk and tape software. Call Steve at 694-2443 anytime.

FOR SALE: 1 SV-902 disk drive & controller card \$200.00, 1 centronic cable for SV-328 \$20.00, 1 Music Mentor cartridge \$15.00, 1 copy of Les Dessous du SV-318/328 \$10.00 and 1 copy of Introduction to Basic \$7.00. All material is used but in good condition. All prices firm and all sales final. For more information call (514)455-4705 anytime.

FOR SALE: 1 SV-328 computer, 1 SV-601 expander, 1 80 column card, 1 centronics card, 1 modified double side double density disk controller card and 2 double sided double density drives. Call Claude weekdays from 9-5 at 843-3448.

FOR SALE: 1 SV-328 computer, 1 SV-601 expander, 1 printer card and 1 Spectravideo printer. Call Yves after 6pm at 626-0014.

Si la reunion de septembre se deroule bien, la prochaine se tiendra le 1er Octobre a 19h.30, au 3460 Peel, #111.

NOUVELLES SMUG

Mr Uwe Schroeder, du groupe d'utilisateurs SVI/MSX d'Allemagne de l'Ouest, nous informe que le distributeur allemand de SVI vient de laisser tomber les ordinateurs SV-318 et 328. Nous aimerions remercier Mr. Schroeder pour les programmes du domaine public en SV-Basic qu'il nous a fait parvenir. Il nous a aussi envoye 6 numeros du magazine allemand appele "MSX Revue" qui semble tres interessant, meme si je ne comprends pas l'Allemand. Cependant, si vous le lisez et cherchez a vous y abonner, vous pouvez ecrire a cette adresse pour plus de renseignements:

MSX REVUE
Postfach 1107
8044 Unterschleissheim
Allemagne de l'Ouest

Encore des nouvelles du groupe d'Allemagne! Douglas Boyd, un membre du groupe a offert de nous traduire certains des articles qui paraissent dans leur bulletin.

M. Wilhelm Pitzeier Jr. du club autrichien SV nous a recemment contactes pour nous demander si nous etions interessees a echanger notre bulletin et des informations. Naturellement, nous l'etions! Nous devrions recevoir une copie de leur bulletin tres bientot.

A propos de bulletin, Mr. Schroeder nous a raconte qu'apres la premiere reunion de leur groupe en mai, la plupart des numeros de notre bulletin qu'il avait recus, avaient disparu. Il suppose que les membres presents les auraient subtilises.

Computers Distributors Limited, de Nouvelle Zelande, vient de fermer L'aide qu'ils nous apportaient nous manquera beaucoup.

Le bruit court a l'effet que MSX User soit encore publie. Je suggere donc aux membres qui s'y sont abonnees mais n'ont jamais rien recu d'ecrire a l'adresse suivante et de s'occuper de leurs affaires:

ARGUS SPECIALIST PUBLICATIONS LIMITED
No. 1 Golden Square
London W1R 3AB
England

Si Argus ne donne rien, l'agence suivante peut vous aider soit a rentrer dans votre argent, soit a recevoir votre annee de MSX User:

THE ADVERTISING STANDARDS AUTHORITY
Dept 2, Brook House
Torrington Place, London WC1E 7HN
England

Apres tout, ils ont utilise votre argent pendant une annee entiere sans rien vous donner en echange. Que coutent quelques minutes de votre temps et 68 cents de timbres compares aux \$30.00 que vous avez deja investi?

LE CONCEPT DE L'ORDINATEUR SYMBIOTIQUE

Par John Durham

(Une extension de la maniere dont fonctionnent les ordinateurs)

La plupart des ordinateurs modernes sont concus autour du concept de l' unite de traitement unique. Les petits systemes utilisent un microprocesseur unique, tandis que les plus importants varient du gros

processeur unique a plusieurs petits processeurs travaillant autour d'une memoire globale. Dans la plupart des systemes, des memoires intermediaires sont installees pour les proteger des lourdes charges de sortance.

La premiere etape pour changer cet etat de choses est d'introduire un deuxieme ensemble de memoire tampon dans la region entre l' unite centrale de traitement et sa memoire principale. La memoire originale accomplira toujours ses fonctions habituelles qui consistent a diriger les lignes d'adresses et de donnees qui controlent le systeme de memoire interne principal, les portes, etc. Le second ensemble est utilise pour prendre controle du bus de systeme d'un ordinateur a l'exterieur de l'hote, juste assez longtemps pour accomplir le transfert de donnees vers le systeme de memoire externe.

LE MECANISME DE CONTROLE:

Il est regle par une porte situee entre l'Unite Centrale (CPU) et ses tampons principaux. Le but de cette porte est de diviser la memoire disponible en quatre fenetres identiques representees, chacune par un octet dans la porte. L'etat de l'octet determine si une fenetre-memoire existe a l'interieur ou a l'exterieur du systeme. Tous les octets sont fixes a zero a l'allumage, de maniere a assurer, pendant l'amorçage, que toute memoire sera interne au systeme et que seuls les ensembles de tampons conventionnels seront utilises. Si l'un de ces octets est fixe a 1, et que l' unite centrale s'adresse a cette partie de la memoire, le deuxieme ensemble de tampons est utilise pour demander l'accès a la memoire exterieure en vue de cette operation.

Pendant les operations exterieures memes, l'accès est controle par un systeme de requete et d'accuse de reception par bus qui permet aux lignes du bus non utilisees dans le systeme externe d'etre pretees brievement. L'Unite Centrale se sert alors la memoire externe de la meme facon qu'elle utilise sa propre memoire. Cela veut dire que de gros blocs de donnees peuvent etre deplaces de la memoire d'un systeme a celle d'un autre en utilisant des instructions directes comme LDIR pour le Z80. L'Unite Centrale n'a aucune connaissance de ce processus car il parait se faire a l'interieur de son propre espace de memoire.

Les operations externes elles-memes sont controlees par un reseau de circuits logiques qui utilise les donnees disponibles sur les portes internes pour decider de quel ensemble de tampons il se servira, ou si le masquage de ROM ou de RAM devra prendre place. Un octet est aussi fourni sur la porte interne qui controle si le ROM d'amorçage est en operation ou non, ainsi qu'un autre octet pour controler si les portes logiques sont a l'interieur ou a l'exterieur du systeme (Il peut utiliser les peripheriques de son hote comme si c'etaient les siens, le cas echeant)

Dans cette configuration, le systeme principal et son systeme externe (hote) existent dans une relation de symbiose, dans laquelle chacun peut accomplir ses propres operations, mais les donnees passent dans les deux sens sous le controle du systeme principal, directement d'une memoire a une autre. Ainsi, en creant un protocole de communication, le systeme hote peut laisser des messages qui sont ramasses par le systeme principal, et le systeme principal envoie ses messages directement a son hote.

ARBITRAGE DE BUS:

Par definition, le reseau de circuits logiques est l'arbitre de l'usage fait de la memoire de l'hote, par le systeme principal et par l'hote. Cependant, comme les bus des deux systemes sont separes entre les systemes internes et externes, la logique pourrait donner acces au bus interne du systeme principal pendant que l' unite de controle accede a l'externe. De cette maniere, le systeme principal peut devenir l'hote d'une

autre machine identique qui serait connectée à son propre bus et système de mémoire internes. Par requêtes et accusés de réception conventionnels, le système principal peut donner accès à une autre unité qui l'utiliserait comme hôte. Grâce à ce petit tour de passe-passe, une unité d'ordinateur est formée qui peut s'étendre de système en système pour former un arbre binaire d'Unités Centrales presque à l'infini.

IMITATION DU CERVEAU:

La raison pour laquelle les systèmes sont conçus ainsi a rapport avec les différences qui existent entre les systèmes d'ordinateurs modernes et le cerveau. Les ordinateurs dépendent d'un processeur qui accomplit toutes les opérations en séquence, comme spécifié par le programme. Le cerveau humain, d'autre part, bien que de loin plus lent en opération, peut manipuler des problèmes plus complexes, en utilisant plus d'information, à vitesse de l'éclair. Ceci parce que le cerveau possède des milliers de canaux qui opèrent sur le même problème en même temps, et assez de capacité pour travailler sur d'autres choses en même temps. Ce concept d'ordinateur, s'étendant en un arbre binaire de taille énorme, accomplit une sorte de simulation de cet effet en permettant à plusieurs Unités Centrales de travailler ensemble ou séparément, dépendant des circonstances.

COMPARAISONS AVEC LES SYSTEMES CONVENTIONNELS:

Travailler avec un tel système apporterait un certain nombre d'avantages et de désavantages:

AVANTAGES:

1: Toutes les unités du système peuvent être virtuellement identiques, sauf les périphériques, ce qui les rendrait faciles à concevoir et de fabrication bon marché.

2: Le transfert de messages par mouvement parallèle de blocs de mémoire est le moyen le plus rapide de le faire. Les machines seraient donc très rapides.

3: Elles pourraient être fabriquées avec un bus standard identique, ce qui rendrait les périphériques interchangeables.

4: Presque toute Unité Centrale capable de se servir de portes pourrait être utilisée dans ce système.

5: Le système peut être étendu à l'infini.

6: Le système est capable de traiter des applications à tâches multiples de haut grade, grâce à ses nombreuses Unités Centrales et sa possibilité de transfert parallèle de données.

7: La relation de symbiose entre les unités locales, qui donne son nom au concept, permet à chaque Unité Centrale d'opérer au double de sa capacité normale, de portes et de périphériques.

8: Ce système serait capable de contrôler des activités quasi-humaines, comme l'ouïe, la parole, la vision, le mouvement, un haut niveau de décision, et l'aménagement interne, le tout simultanément.

9: Il est idéal pour usage dans le style androïde et robots industriels.

10: Parce que son bus peut être assimilé à celui d'un ordinateur de maison, il peut être programmé par lui et le traite comme son hôte initial. Les langages conventionnels peuvent être utilisés.

11: La technologie existante est capable de bâtir les unités. Aucun composant spécial ou de vitesse ultra-rapide n'est nécessaire.

12: Chaque Unité Centrale ne traite qu'avec un seul hôte et deux périphériques qui reçoivent les messages ou les lui envoient.

DESAVANTAGES:

1: La taille relativement petite de la mémoire dans chaque Unité Centrale, optimisant l'efficacité de l'unité, rend difficile à manipuler

l'operation des bases de donnees importantes, a cause du montant limite d'information que chaque Unite Centrale peut controler. Les limites de cette sorte, cependant, sont solutionnees par l'alphabetisation des entrees des bases de donnees, et le partage entre les differentes Unites centrales des differentes parties des listes importantes. Ainsi, les bases de donnees et les recherches par listes seraient rendues plus rapides.

2: Le potentiel de ce systeme a s'etendre pourrait mener a de gros systemes exigeant de multiples ou importantes sources d'approvisionnement energetiques. Ce probleme existe cependant deja dans les ordinateurs conventionnels de grosse taille. Ce probleme pourrait par contre etre soulage par l'usage de pieces CMOS ou HCMOS dans sa construction, ce qui exigerait bien moins d'energie que les LSTTL.

3: Dans de tels systemes, la multiplication des circuits imprimes augmenterait le volume des appareils, limitant demesurement leur mobilite, comme dans le cas de robots. Ceci pourrait etre solutionne par la creation de circuits qui comprendraient plusieurs unites d'ordinateurs sur une meme surface, augmentant ainsi la puissance et reduisant d'autant les problemes d'interferences auxquels ces systemes sont sujets.

4: Les transferts de donnees en parallele auraient lieu a des vitesses enormes a travers les cables et connecteurs. Le choix de connecteurs a hautes performances, de cables blindes et de chips de memoire de bonne qualite, la reduction des sauts de pistes et l'emploi de condensateurs pour reduire les bruits d'interference feraient partie des considerations essentielles durant la conception. Cette fonction exige de plus que les cables-rubans soient courts, puisque leur fiabilite decroit au carre inverse de la distance, a partir de la source du signal.

5: Les communications a longue distance se feraient toujours sur lignes serieelles, permettant que soient maintenus les interfaces RS232 et liaisons en boucles. Ceci, pourtant n'est pas essentiel, puisque beaucoup de systemes seraient capables d'operer sans lignes de donnees ou d'acquies les donnees d'autres sources.

6: Certains groupes comme les utilisateurs commerçants pourraient etre enclins a critiquer ce genre de developpement a cause du manque d'applications immediates evidentes ou operations rentables en commerce. Les applications commerciales sont bien servies par les systemes d'ordinateurs modernes, et ce systeme particulier ne devrait pas etre considere comme le fin du fin en matiere de developpement informatique. Il est concu pour des applications specialisees, valables pour des systemes intelligents, l'education heuristique, la robotique et les applications generales exigeant des machines intelligentes, non pas pour le commerce.

ENVIRONNEMENT:

Pour l'Unite Centrale, il n'y a pas de distinction entre la memoire qui se trouve a l'interieur et celle qui est a l'exterieur du systeme. Aucune distinction non plus entre les peripheriques controles par ses propres portes et ceux de son voisin immediat. Si le systeme possede les programmes permettant de les operer, ils fonctionneront aussi bien. Certains peripheriques qui interrompent leur hôte peuvent etre programmes par une machine externe de maniere a annoncer les messages qui arrivent, par exemple.

Les ROM du systeme d'exploitation pourraient etre mis en action ou stoppes, utilisant ainsi au maximum la memoire disponible en RAM. On suggere que la memoire sous controle direct de l'Unite Centrale soit divisee en deux blocs. L'un d'eux peut etre utilise pour le systeme d'exploitation, mis en fenetre pour les echanges de donnees entre systemes et pour l'amenagement general. L'autre moitie est alors mise de cote pour

l'utilisation de programmes qui peuvent fonctionner sans interference, sans etre forces de s'ecraser en cas d'accès inattendu par les autres systemes.
SYSTEMES D'EXPLOITATION AUTONOMES:

Ce terme a ete invente pour s'adapter a ce genre de machines. Il decrit un systeme d'exploitation en code machine, qui actionne et teste les diverses unites, garde le controle du reseau d'Unites Centrales, manipule le transfert de toutes les donnees entre systemes, et execute les programmes, controle les memoires et les fenetres, les peripheriques, la distribution intelligente du travail et des taches invisibles et adroites qu'on attend d'un tel systeme. Ces systemes dependent enormement de leur equipement, et devraient tenir compte de leur hôte et des programmes qui y travailleront. Dans une certaine mesure, l'hôte devra emuler l'ordinateur de depart, de maniere a accepter les programmes deja developpes, autant que possible sans modification.

Par exemple, supposons qu'une Unite soit connectee a un hôte System 80. Le systeme d'exploitation pourrait choisir de detourner les 16 premiers K de memoire vers les operations externes, faisant siens le ROM, le clavier et l'ecran du System 80, puis amorcer le systeme d'exploitation du System 80 a partir des drives, acceder un programme et repasser aux operations internes, liberant ainsi le System 80 pour usage normal. Pendant ce temps, le System 80 continuerait sur sa lancée sans savoir que son equipement est utilise par un systeme externe. Le systeme externe pourrait etre programme pour detourner temporairement les routines de ces peripheriques vers une boucle inoffensive pendant que l'accession est accomplie, puis les restorer quand l'accession est terminee.

Quiconque est interesse a discuter ce developpement est invite a ecrire a

JOHN DURHAM
16 Hudson Ave.
Upper Hutt
New Zealand

DIMENSIONNEMENT DE LA FENETRE ECRAN EN MODE TEXTE Par: Denis Landry

```

      ECRAN
0-----! 233
2!      ! 234
4!      ! 235
!      !
10!     !
!      !
!      !
24!-----! 254

```

3 POKE 64006!,0

5 CLS

10 SCREEN 0,0 : 'ENLEVE LA DERNIERE LIGNE (KEY)

20 LOCATE 0,23: 'POSITIONNEMENT

30 PRINT " 1 - ADDITION 2 - SOUSTRACTION ": ' NOTRE MENU

50 LOCATE 0,0 : ' REPOSITIONNEMENT EN DEBUT D'ECRAN

60 POKE 64006!,238 : 'POKE QUI DELIMITERA NOTRE ECRAN

70 INPUT "VOTRE CHOIX : ";C

80 IF C=1 THEN GOTO 500 ELSE 1000

500 FOR T=1 TO 25

```

510 PRINT " ";T;" + ";T;" = ";T+T : FOR Q=1 TO 90:NEXT Q
520 NEXT T
550 GOTO 3
1000 FOR T=1 TO 25
1010 PRINT " ";T;" - ";T;" = ";T-T : FOR Q=1 TO 90:NEXT Q
1020 NEXT T
1050 GOTO 3

```

Après un essai ou deux, faite un <CTRL STOP>, faite un <LIST>, vous voyez maintenant une page écran de 5 lignes pour revenir à une page complète faite < POKE 64006,0 >.

PROGRAMMER EN ASSEMBLEUR Part 4 Par: Claude Morissette

ARITHMETIQUE ET LOGIQUE:

On sait déjà que seul l'accumulateur A peut servir pour les opérations logiques, OU, ET, OU-Exclusif, et les comparaisons. Plus haut on avait vu: OUA, qui est en fait A OU A; comme le A est toujours impliqué dans ces instructions, inutile de l'écrire. Ainsi, toutes ces opérations ont la forme-type:

OUR ; &R ; OXR ; R= A,B,C,D,E,H,L ou M la mémoire.

Les comparaisons forme-type: CPR, tout simplement. Ou les memes opérations avec un nombre spécifique, n, 1 byte:

OU n ; & n ; OX n et CP n.

Peu à retenir dans notre "RAM" pour 36 instructions type.

Les forme-types pour arithmétique de 8 bits, 1 byte, TOUJOURS avec l'accumulateur A :

+R , -R ; opérations simples, sans la retenue, "carry".

+R , -R ; le (') signifie opérations avec la retenue, s'il y en a une, provenant d'une opération précédente.

+ n, - n, + ' n et - ' n ; avec n=1 byte spécifique. Un autre 36 instr. pour cette effort!

Pour l'arithmétique de 16 bits, (2 bytes), on sait que HL doit être l'accumulateur, contenant une des valeurs, puis le résultat. Sont exécutables:

+PP ; addition "simple", PP=BC, DE, HL ou SP seulement.

Il n'y a pas de -PP, i.e.e. sans retenue, par conséquent, l'assembleur lit: -'PP ou -PP, comme la même instruction. Mais différentie entre; +PP ; avec retenue (carry), et +PP ; addition sans la retenue. Il n'y a pas, non plus, sur 16 bits, de + ou - nn, 2 bytes qui seraient spécifiés. Les valeurs doivent être dans les registres.

Bien sûr, IX ou IY peut servir d'accumulateur pour addition sans retenue seulement. Et l'exécution est plus lente aussi.

Multiplication et division..., mais, mais, mais que croyez-vous? Les microprocesseurs qui acceptent ces instructions sont rares et plus récents que le Z80. Pourtant, le Basic et d'autres langages le font habilement avec Z80 ou 8080 encore plus ancien. Oui, avec des sous-routines appropriées. En assembleur, on travaille directement avec le microprocesseur et ses limites.

Celui qui veut développer des programmes efficaces, rapides, occupant relativement peu de mémoire, en assembleur, prépare donc ses propres sous-routines de multiplication et division. Faites une fois et vérifiées, elles peuvent toujours être utilisées dans plusieurs programmes. Et très simplement si vous avez un assembleur qui peut trouver sur disque, en librairie, le ou les sous-programmes que vous desirez inclure dans un nouveau programme, sans l'écrire encore.

<<< PLEASE TAKE NOTE / S.V.P. PRENEZ EN NOTE >>>

=====

We are moving and as of the 15th of September our new address will be:

Nous demanageons le 15 Septembre et notre nouvelle adresse sera:

SMUG OF MONTREAL
20 - 159e AVE.
N.D. Ile Perrot, Que.
Canada J7V 5V6

LAST MINUTE NEWS BULLETIN

We have finally recieved issue no. 6 of Micros MSX from France. They will be producing one more issue under the Micros MSX name and then they will be changing the name to Micros ID (Informatique Domestique) to include other popular brands of computers in the same publication along with the MSX mark. All of those who subscribed to Standard MSX or Micros MSX will automatically have the remainder of thier subscriptions transferred to Micros ID.

We would like to thank our friend Graham Wolfe for the mention of our group in his column "Wolfe Bytes" in the Chronicle on August 20th.

The SVI/MSX Club Austria has sent us a recent copy of thier newsletter which is written in German. We'd better start to take German lessons soon with the amount of German publications that we're recieving, so that we'll be able to understand them.

BAS of Norway sent us 3 SV-BASIC public domain disks to add to our library. They have a lot of interesting looking games and utilities on them. Steve Verreault is currently trying to see how they work and what the object of the programs are, so that he can try to rewrite French or English instructions for them.

Last but not least we have recently recieved Volume 1 No. 4 of MSX and Spectravideo Computer Forum, a magazine published in the English language by Australians. SMUG of Montreal was mentionned in this issue as well. It is a magazine strictly dedicated to helping SVI & MSX users make the most of thier computers and other peripherals. The articles are interesting and informative. Canadian users may subscribe by sending \$45.00 U.S. for airmail rates or \$27.00 U.S. for surface mail rates to:

COMPUTER FORUM
Subscription Dept.
Freeport 3, P.O. Box 71
Palmyra, W.A. 6157
Australia

These subscription rates allows you 12 issues of the magazine. For your convenience we will be including a subscription form to COMPUTER FORUM in next month's issue of the newsletter. The address and rates are published here for those of you who may be to anxious to wait.

SMUG of Montreal
20 - 159e Ave.
N.D. Ile Perrot, Que., Canada J7V 5V6

We are the official Spectravideo and MSX user's group for the Montreal area as recognized by Spectravideo Canada, Yamaha Canada and Bondwell International. A family membership entitles you and your family to attend groups meetings, trade programs and receive 1 copy monthly of our newsletter per family. An Individual membership is for individuals who wish to attend the groups meetings, trade programs and receive a copy of our monthly newsletter. Foreign members are allowed to trade or purchase programs and receive monthly newsletters through the mail.

As a member of the group you will be able to purchase Spectravideo hardware and software at reduced prices from SMUG of Montreal. Non-members are permitted to attend the monthly groups meetings, by paying \$3.00 admission fee at the door.

Nous sommes le groupe officiel pour Spectravideo et MSX de la region de Montreal, tel que reconnu par Spectravideo Canada, Yamaha Canada, et Bondwell International. Les familles membres peuvent assister aux reunions du groupe, echanger des programmes et recevoir une copie du bulletin mensuel. Un membre individuel peut aussi assister aux reunions, echanger des programmes et recevoir notre bulletin. Les membres etrangers peuvent echanger ou acheter les programmes aux prix du groupe, et recevoir le bulletin par courrier.

Comme membre du groupe, vous serez autorise a acheter du materiel Spectravideo a prix reduits, chez Smug de Montreal. Les non-membres peuvent assister aux reunions de groupe en payant l'admission de \$3.00 a la porte.

MEMBERSHIP APPLICATION / APPLICATION POUR MEMBRE

Nom/Name _____ Telephone _____

Adresse/Address _____

Ville/City _____ Province _____

Pays/Country _____ Code Postal/Postal Code _____

Modele d'ordinateur/Computer make & model _____

Famille/Family \$30.00(CAN.)	Individuel/Individual \$20.00(CAN.)
U.S.A. \$25.00(U.S.)	Outre-Mer/Overseas \$30.00(U.S.)

Cheque Certifie ou Mandat/ Certified Cheque or Money Order - Payable to:
Jean-Guy Fontaine, 20 - 159e Ave., N.D. Ile Perrot, Que., Canada J7V 5V6

OFFICE USE ONLY - DO NOT WRITE IN SPACE BELOW.

This is to certify that the above member has paid his dues in full and is in good standing as a member of SMUG of Montreal until;

Expiration date _____ Signed and approved by _____



SMUG DE MONTREAL
443 BOUL. KOCHÉ
VAUDREUIL, QUÉBEC
J7V 2N4

Air Mail Par avion

Don Stanley
P.O. Box 7442
Wellington North
New Zealand

PRINTED MATTER

